V R Vasudevan

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EDUCATION

Georgia Institute of Technology, Atlanta, USA **Aug' 2026 – May 2028**
Master of Science in Robotics

Delhi Technological University, New Delhi, India **Dec' 2021 – Jul' 2025**
Bachelor of Technology in Production and Industrial Engineering

TECHNICAL SKILLS

Languages: Python, C++, Bash
Simulation: IsaacLab, Isaac Sim, MuJoCo, Gazebo
Tools & frameworks: ROS, ROS2, PyTorch, JAX, RSL-RL, Stable-Baselines3, OpenVLA, $\pi_{0.5}$, LIBERO, ONNX, Docker, Git, Linux
Hardware: NVIDIA Jetson, Pixhawk, Intel RealSense, Arduino, RaspberryPi

PROFESSIONAL EXPERIENCE

LIDAR LAB, Georgia Tech **Aug' 2026 – Present**
Graduate Researcher | Advisor: Prof. Ye Zhao *Atlanta, USA*

- **Skill Chaining for Humanoid Locomotion:** Training teacher perceptive-locomotion policies for **Unitree R1** for skills such as stair climbing, crouching, etc. Leveraged **Attention-based map encoder** to compress height maps and PPO with **privelaged critic** to train the policy.
- Increased the robustness of the policy by introducing an adaptive stair curriculum rooted in **shaped rewards** to align skill gait and style with retargeted motion priors. Achieved waypoint RMSE of 0.1 m, scaling step heights between 10-25 cm in simulation for stair skill.

Robotics Research Center, IIIT-Hyderabad **Nov' 2025 – May' 2026**
Research Internship | Advisor: Prof. Spandan Roy *Hyderabad, India*

- **ML for Aerial Robotics:** Trained a control policy for quadrotor position control with **PPO** in IsaacSim, achieving hover RMSE of 2 cm in simulation and 10 cm in real. Exported to **ONNX** for deployment on Jetson Orin Nano at 100 Hz, successfully executing **Sim2Real**.
- Trained an adaptable end-to-end neural network controller for motor control by implicit model estimation based on state-action pair history, helping reduce the Sim2Real gap.
- Deployed **diffusion policies** onboard a multirotor platofrom for estimating acceleration residuals integrated into an **Adaptive Sliding Mode Controller (ASMC)** to improve control robustness under changing inertia, such as in aerial manipulation.

Team Unmanned Aerial Systems-DTU (UAS-DTU) **Feb' 2022 – Jul' 2025**
Avionics Head (previously engineer) | Advisor: Prof. Indu Sreedevi *Delhi, India*

- Led avionics R&D for a 20+ member team in perception, planning, and control; team podiumed at 3 international competitions, including 2nd among self-funded teams at **DARPA Triage Event-I**.
- Developed a target-following framework for UAVs. Estimated local cartesian position of the target using a calibrated gimbal mounted monocular camera with RMSE 5 m & follow speeds up to 8 m/s. Solution demonstrated on hardware at **IMAV 2023** at Aachen, Germany.

Ecole Polytechnique Federale de Lausanne (EPFL) **Jul' 2024 – Aug' 2024**
Research Internship | Advisor: Prof. Jan Skaloud *Lausanne, Switzerland*

- **OpenSimDrive:** Implemented a **Vehicle Dynamics Model** and integrated it into a **graph-based sensor fusion** framework reducing localization error by 60% compared to IMU-only odometry under GNSS denial, validated on 3 hrs of real-world driving data.

PUBLICATIONS

S. Pratap Singh*, S. Ujjwal*, I. Chaudhary, **V. R. Vasudevan**, R. D. Dev, S. Roy. ***Reinforcement Learning with Inner-Loop Dynamics Estimator for Aerial Manipulation under Uncertainty.*** IEEE CASE 2026.

V. R. Vasudevan, S. Grover, S. Indu. ***UNIFollow: Natural Language Conditioned Target Following Framework for UAVs.*** IROS 2025 Workshop (LeaPRiDE).

A. Srivastava, **V. R. Vasudevan**, Harikesh, R. Nalanthiga, P. B. Sujit. ***A Modified Artificial Potential Field for UAV Collision Avoidance.*** ICUAS 2023.

V. R. Vasudevan, K. J. Paul, J. Skaloud. ***Model-Based Constraint for Trajectory Recovery in Low-Cost UGVs.*** In preparation, Robotics and Autonomous Systems.

PROJECTS

Data Augmentation for Vision Language Action models **Sept' 2026 – present**

- Developing a synthetic data generation technique leveraging robot manipulation video generation and **inverse dynamics models** for retargeting and generating action sequences, based on **Dream2Flow**.
- Set up the **LIBERO benchmark** for data generation and wrote the **finetuning** pipeline for OpenVLA and pi0.5. Set up multi-GPU training pipeline on the Georgia Tech compute cluster.

DARPA Triage Challenge **Nov' 2023 – Oct' 2024**

- Responsible for the development of a **UAV-UGV hybrid system** consisting of 4 agents to perform automated triaging in **mass casualty incidents (MCI)**. Assisted in rover design by introducing a cost-efficient platform for efficient sensor orientation.
- Contributed to **mission planning** to enable UxVs to operate efficiently, including automated vital sign assessment upon casualty detection. Part of the final crew presenting the solution at the **DARPA Triage Challenge Event-I** at Georgia, USA.

HONORS AND AWARDS

- Awarded the **Research Excellence Award** by Delhi Technological University at the Golden Pride Ceremony 2026.
- Placed 2nd among self-funded teams at the **DARPA Triage Challenge Event-I**, winning a prize of **$60,000**.
- Attained 3rd position in the **International Conference on Unmanned Aircraft Systems UAV Challenge 2024**.
- Won 3rd position in the **IMAV Hiker Rescue Challenge 2023**, held at RWTH University, Germany.